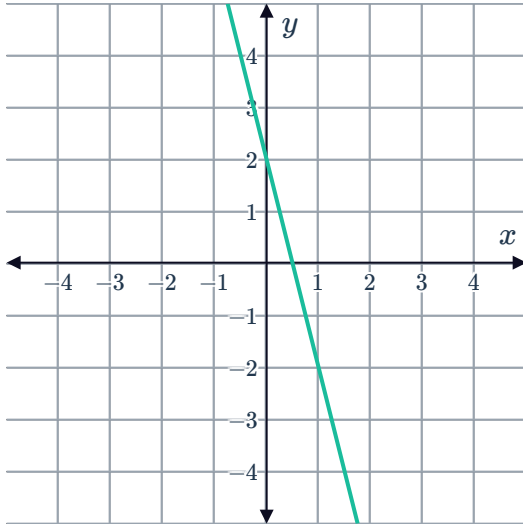


# Transformations of $y=x$

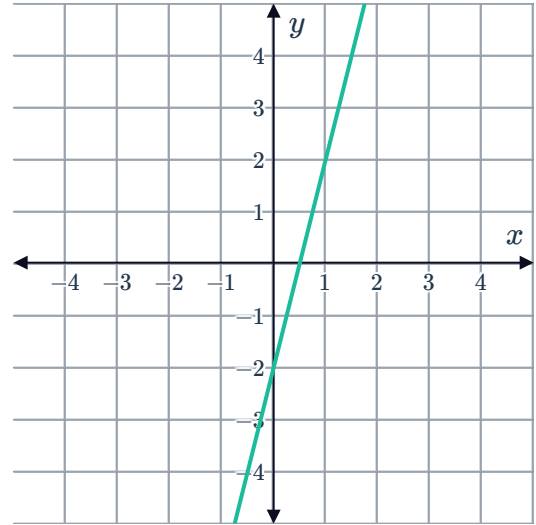


1 Determine whether the following graphs are decreasing linear function.

a

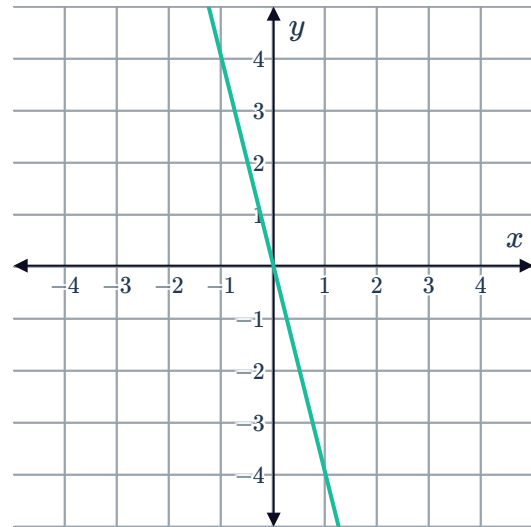


b



2 Consider the graph of  $y = -4x$ .

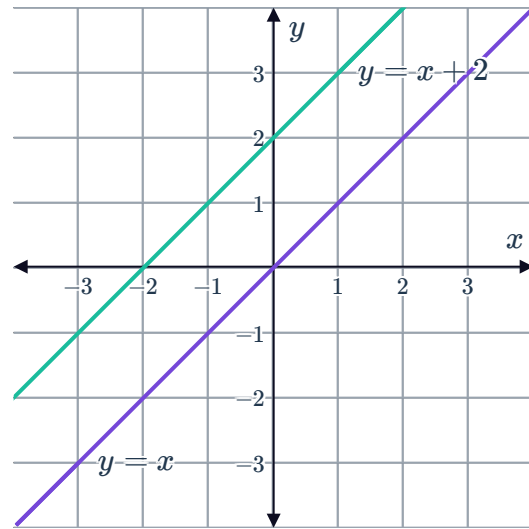
- a As  $x$  becomes larger in the positive direction (i.e.  $x$  approaches  $\infty$ ), what happens to the corresponding  $y$ -values?
- b As  $x$  becomes larger in the negative direction (i.e.  $x$  approaches  $-\infty$ ), what happens to the corresponding  $y$ -values?



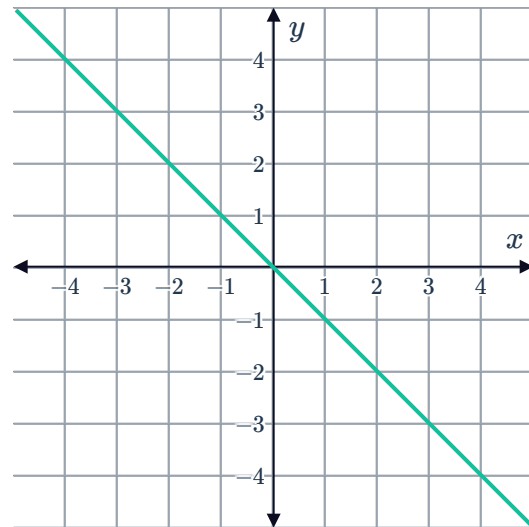
- 3 The lines  $y = x + 2$  and  $y = x$  have been graphed on the same coordinate plane. Determine whether the line  $y = x + 2$  is:



- a Increasing or decreasing.
- b Steeper, flatter or equally steep as  $y = x$ .

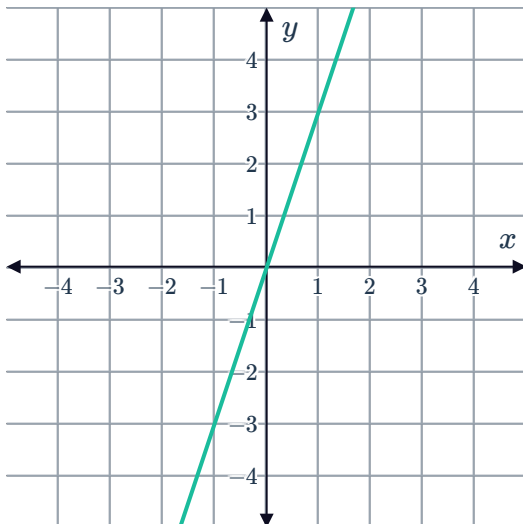


- 4 Consider the given graph of  $y = -x$ .

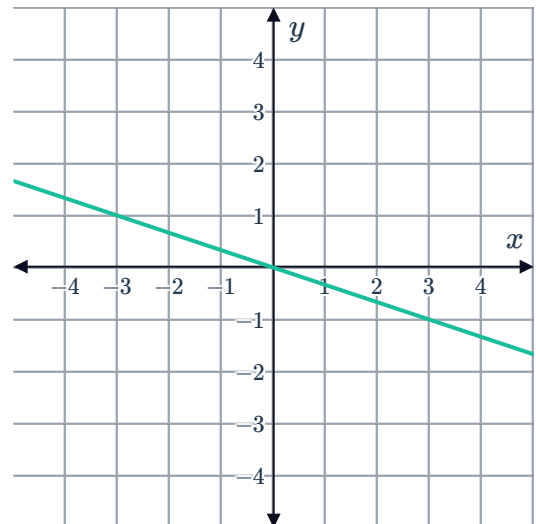


Determine whether the following shows the graph of a straight line whose  $y$  value is decreasing at a faster rate than  $y = -x$ .

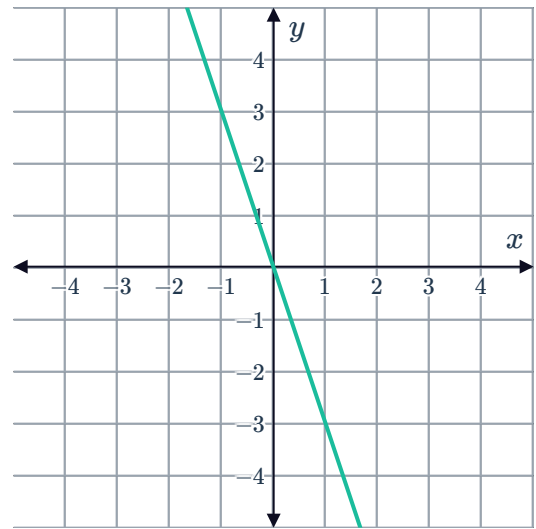
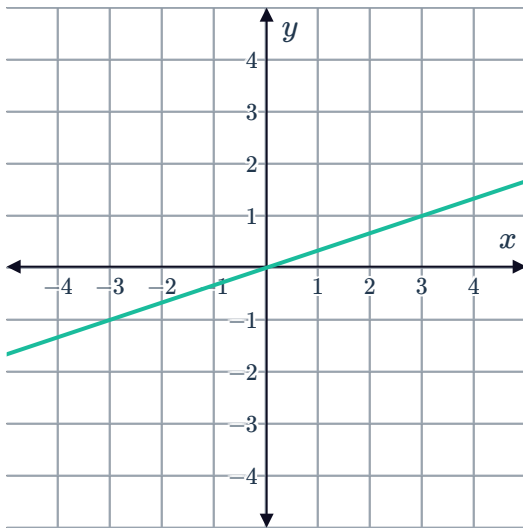
a



b



c

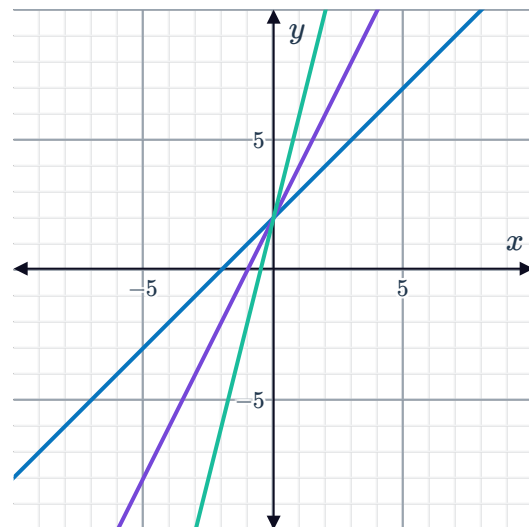


5 A straight line graph has a positive  $y$ -intercept and a positive  $x$ -intercept. Determine whether the following must also be true of the straight line graph.

- a The straight line graph is flatter than  $y = x$ .
- b The straight line graph is increasing.
- c The straight line graph is steeper than  $y = x$ .
- d The straight line graph is decreasing.

6 The equations:  $y = x + 2$ ,  $y = 2x + 2$  and  $y = 4x + 2$  have been graphed on the same number plane:

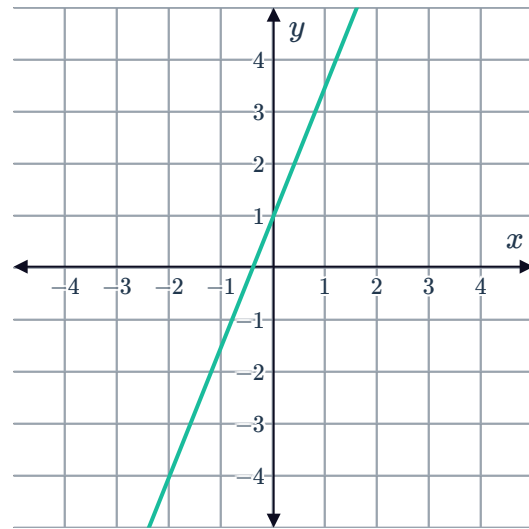
- a What do all of the equations have in common?
- b What do all of the graphs have in common?
- c What can you conclude from the answers above?



7 Look at the graph of the line.



- a At the  $y$ -intercept,  $x = 0$ .  
What is the  $y$ -value at this point?
- b The equation of the line is  $y = \frac{5x}{2} + 1$ .  
For a straight line equation of the form  $y = mx + b$ , which variable represents the  $y$ -intercept?

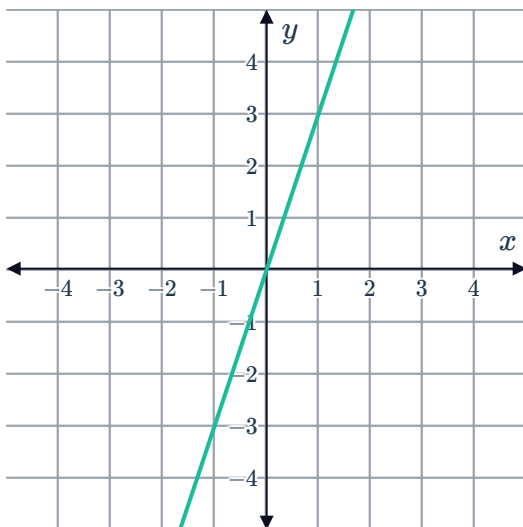


8 Consider the following graphs:

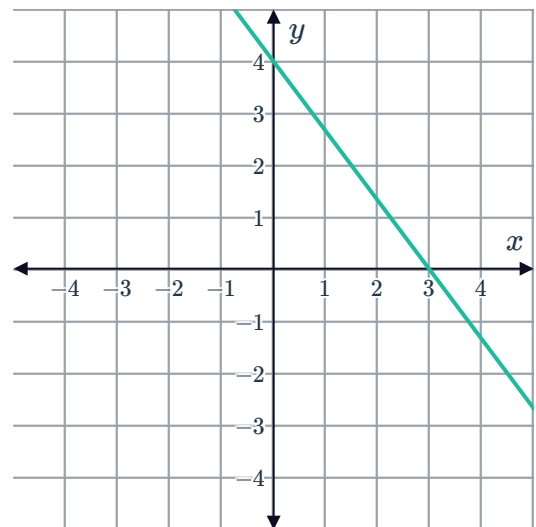
i State the  $x$ -value of the  $x$ -intercept.

ii State the  $y$ -value of the  $y$ -intercept.

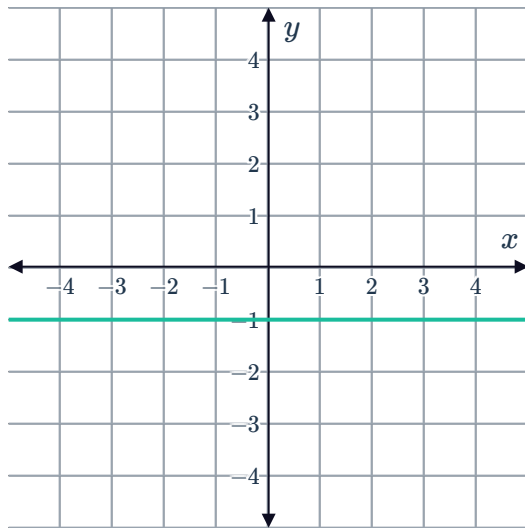
a



b



c



9 Determine whether the following equations represent lines that will intersect the  $y$ -axis at 2.

a  $y = 2x - 5$

b  $y = 2x$

c  $y = 2 - x$

d  $y = 5x + 2$

e  $y = x - 2$

f  $y = x + 2$

10 A linear graph has a negative  $y$ -intercept and a positive slope.  
Is the  $x$ -intercept negative or positive?

11 Determine whether the following lines pass through the origin.

a  $y = 6x$

b  $y = \frac{x}{6}$

c  $y = 6x - 6$

d  $y = 8x + 8$

e  $y = 8x$

12 Find the  $y$ -value of the  $y$ -intercept for the following.

a  $y = -7x + 5$

b  $y = 5x$

c  $y = 10 - 5x$

d  $y = -x + 1$

13 Determine whether the following lines have a  $y$ -intercept.

a  $y = x$

b  $x = 3$

c  $y = -3$

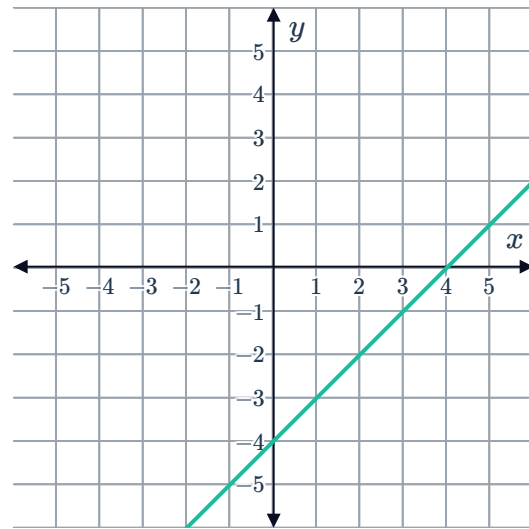
d  $3x - y = 0$

e  $y = 4x + 5$

14 Consider the graph of  $y = x - 4$ .

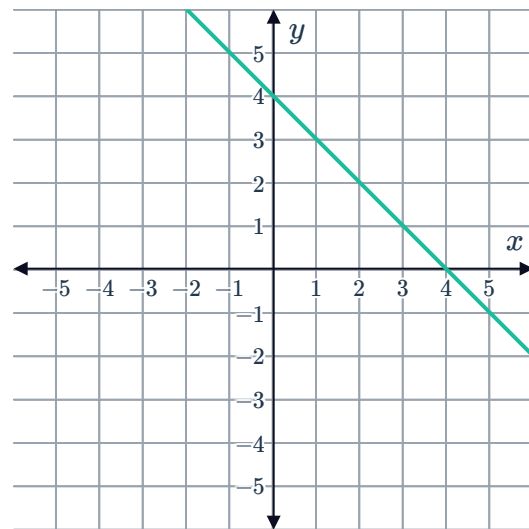


- a State the coordinates of the  $x$ -intercept.
- b State the coordinates of the  $y$ -intercept.
- c If the graph is translated 6 units down, find the coordinates of the new  $y$ -intercept.

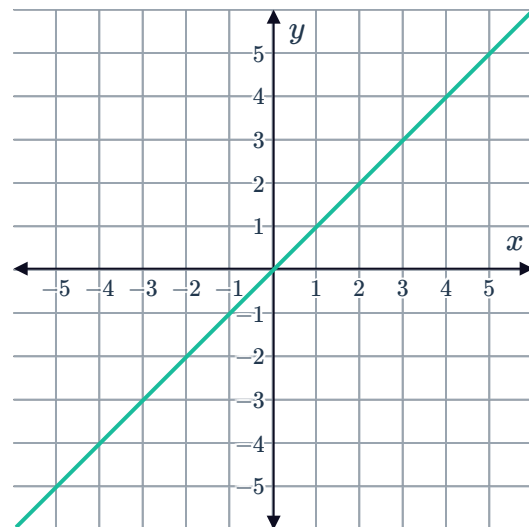


15 Consider the graph of  $y = 4 - x$ .

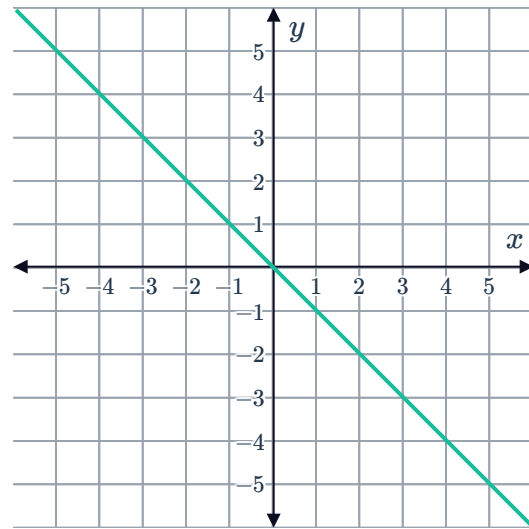
- a State the coordinates of the  $x$ -intercept.
- b State the coordinates of the  $y$ -intercept.
- c If the graph is translated 10 units up, find the coordinates of the new  $y$ -intercept.



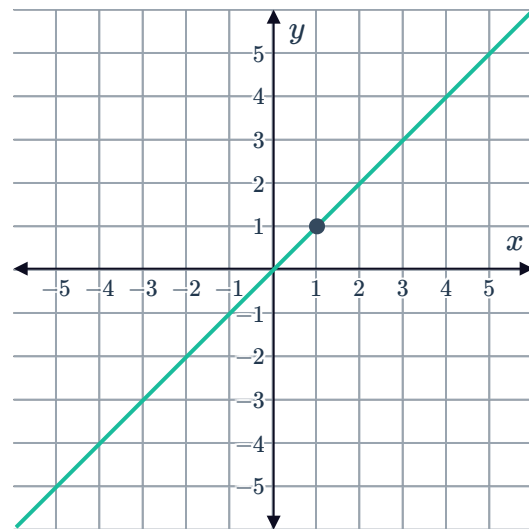
16 Consider the graph of  $y = x$ . Draw the graph of the resulting line translated 2 units down.



- 17 Consider the graph of  $y = -x$ . Draw the graph of the resulting line translated 4 units down.

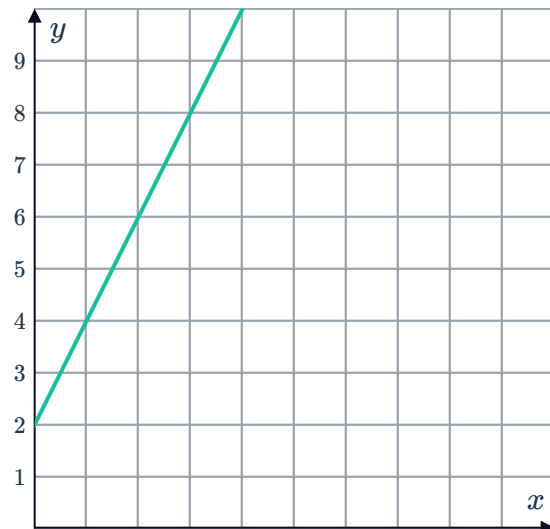


- 18 Consider the graph of  $y = x$ , with the point  $(1, 1)$  that lies on the graph. The steepness of this graph is to be quadrupled. Draw the graph of the resulting line.



- 19 Part of a continuous straight line graph has been graphed. Determine whether each of the following statements is true about the entire line graph.

- a The points on the graph have both positive and negative  $y$  values.
- b The graph has a negative  $x$ -intercept.
- c The graph passes through the origin.
- d The points on the graph have only positive  $y$  values.
- e It has a negative slope.

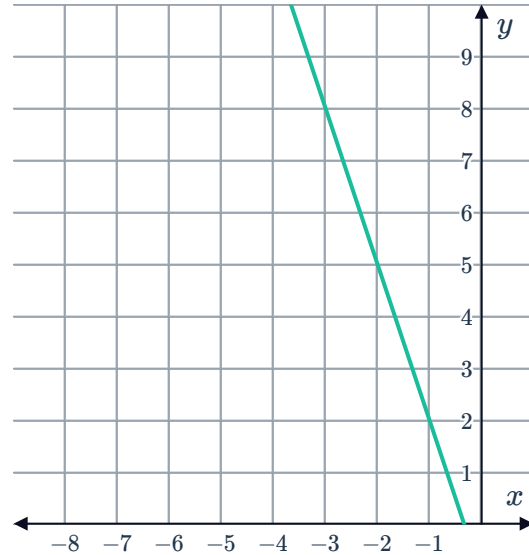




**20** Part of a straight line graph has been graphed.

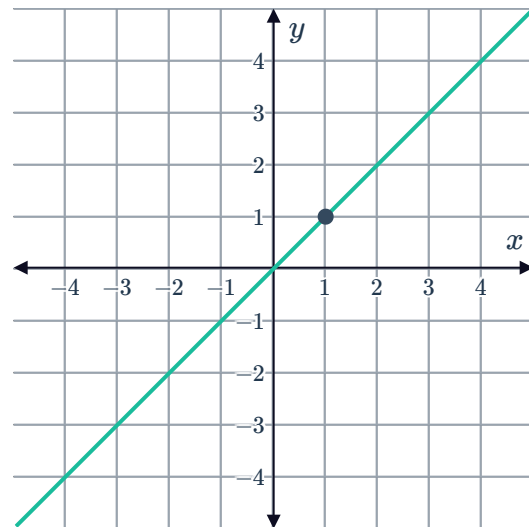
Determine whether each of the following statements is true of the entire line graph.

- a** The points on the graph have both positive and negative  $y$  values.
- b** The points on the graph have only negative  $y$  values.
- c** It has a negative slope.
- d** The graph is increasing.
- e** The graph has a positive  $y$ -intercept.
- f** The graph passes through the origin.



**21** Consider the graph of  $y = x$ .

Where will the point  $(1, 1)$  move to if the line is made 3 times as steep?

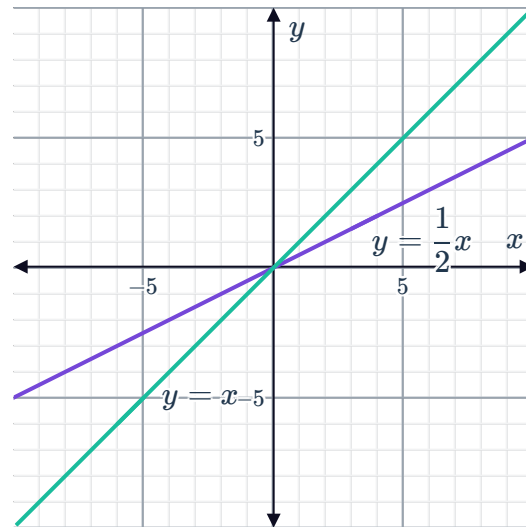




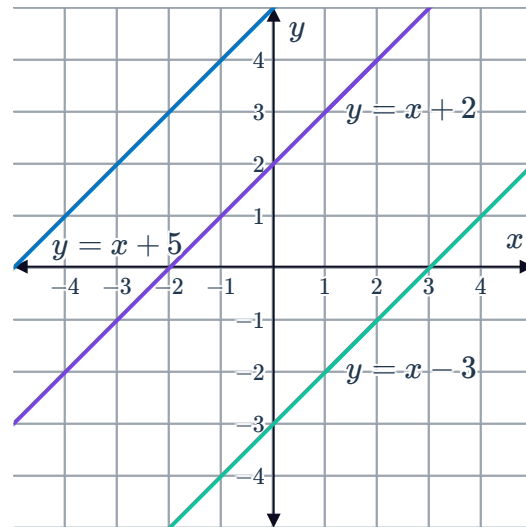
22 The lines  $y = \frac{1}{2}x$  and  $y = x$  have been graphed in the same set of axes.



- a Is the line  $y = \frac{1}{2}x$  increasing or decreasing?
- b Is it steeper, flatter or equally steep as  $y = x$ ?
- c Determine whether the following could be done to transform the graph of  $y = x$  into  $y = \frac{1}{2}x$ .
  - i Reflecting  $y = x$  about the  $x$ -axis.
  - ii Reflecting  $y = x$  about the  $y$ -axis.
  - iii Making  $y = x$  steeper.
  - iv Making  $y = x$  flatter.



23 Consider the graph of the lines  $y = x + 5$ ,  $y = x + 2$  and  $y = x - 3$ .



- a Determine whether the following statements are true.
  - i They are all the same steepness.
  - ii All lines that are parallel to  $y = x$  have positive  $y$ -intercepts.
  - iii All lines that are parallel to  $y = x$  have positive  $x$ -intercepts.
  - iv They can all be made to overlap  $y = x$  through a vertical translation.
  - v They are all parallel to one another.

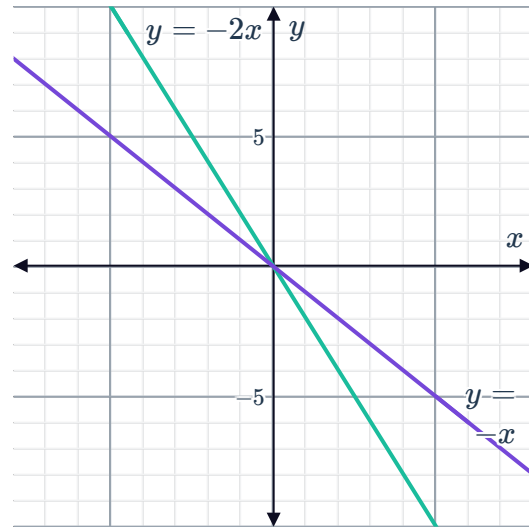
**b** Determine whether the following are true or false about the  $x$ - and  $y$ -intercepts of lines that are parallel to  $y = x$ .



- i** Each line has  $x$  and  $y$  intercepts that are equal.
- ii** The  $x$  and  $y$  intercepts of these lines are the same distance from the origin.
- iii** Each line has  $x$  and  $y$  intercepts that are equal in magnitude, but opposite in sign.

**c** If a straight line is parallel to  $y = x$  and has a  $y$ -intercept of  $-8$ , find its  $x$ -intercept.

**24** The lines  $y = -2x$  and  $y = -x$  have been graphed on the same set of axes.



- a** Is the line  $y = -2x$  increasing or decreasing?
- b** Is it steeper, flatter or equally steep as  $y = -x$ ?
- c** Here are some transformations.
  1. Reflection about the  $x$ -axis.
  2. Steepening of straight line.
  3. Reflection about the  $y$ -axis.
  4. Translation 2 units down.
  5. Flattening of straight line.
  6.  $90^\circ$  rotation about the origin.

Determine whether the following pairs of transformations turn the graph of  $y = x$  into the graph of  $y = -2x$ .

- i** Transformations 1 and 2
- ii** Transformations 2 and 3
- iii** Transformations 1 and 5
- iv** Transformations 2 and 6
- v** Transformations 4 and 5

25 Which of the following statements is true for each of the following set of points.



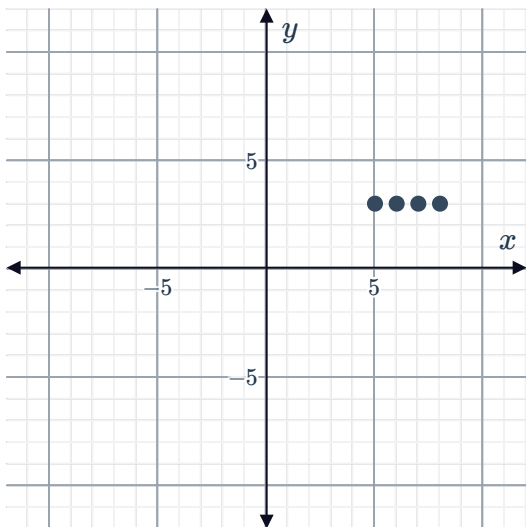
Statement 1: The set of points lie on a vertical line.

Statement 2: The set of points lie on a horizontal line.

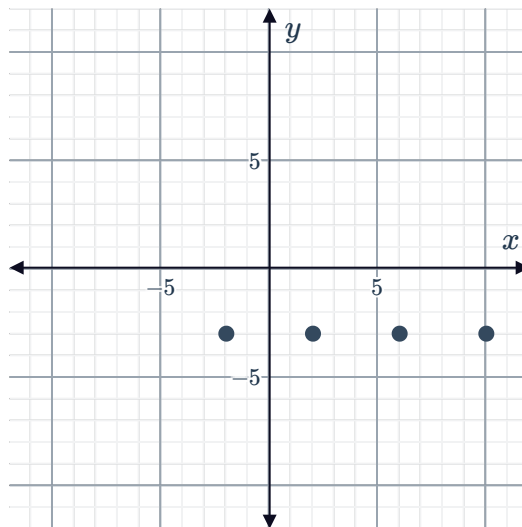
Statement 3: The set of points lie on an increasing line.

Statement 4: The set of points lie on a decreasing line.

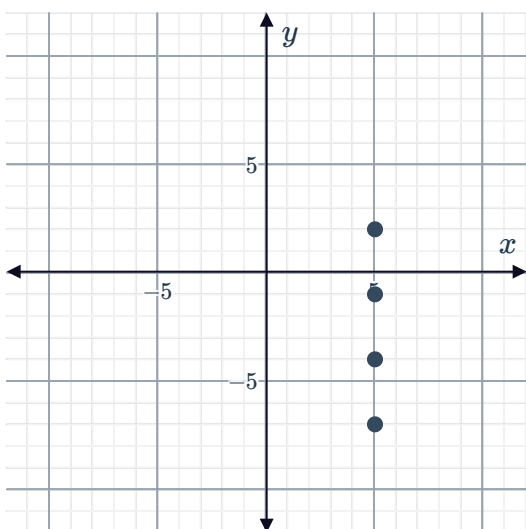
a



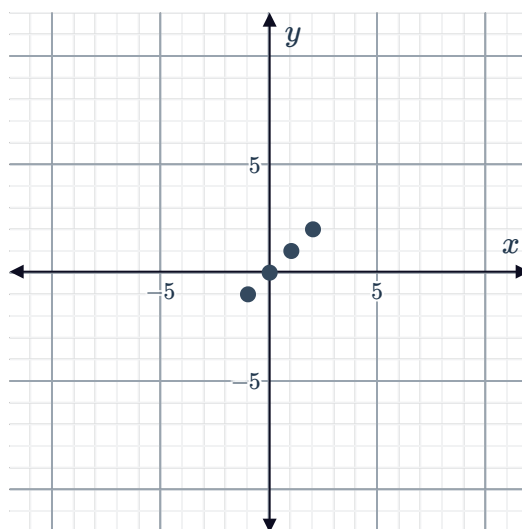
b



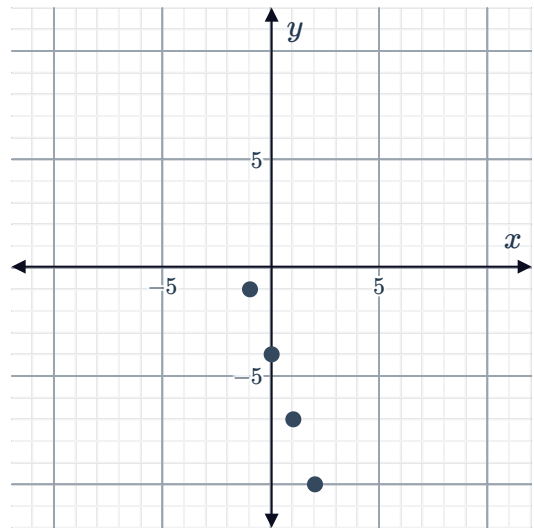
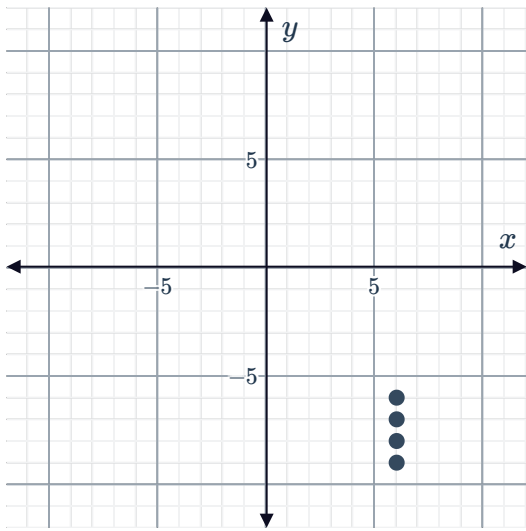
c



d



e



**26** Determine whether the graph of the following are horizontal or vertical line.

**a**  $y = 6$

**b**  $x = -4$

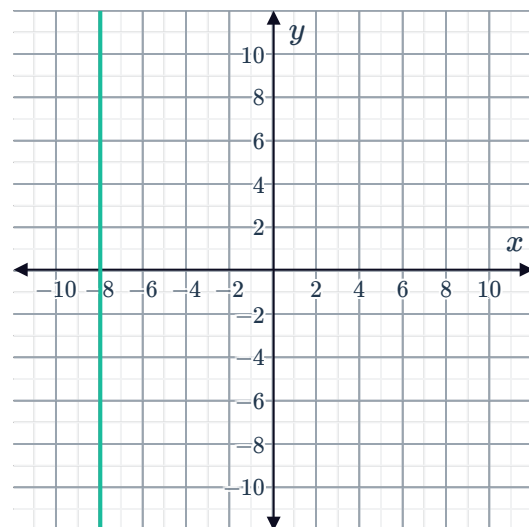
**27** Plot the graph of the following lines:

**a**  $y = -7$

**b**  $x = -8$

**28** Consider the graphed line.

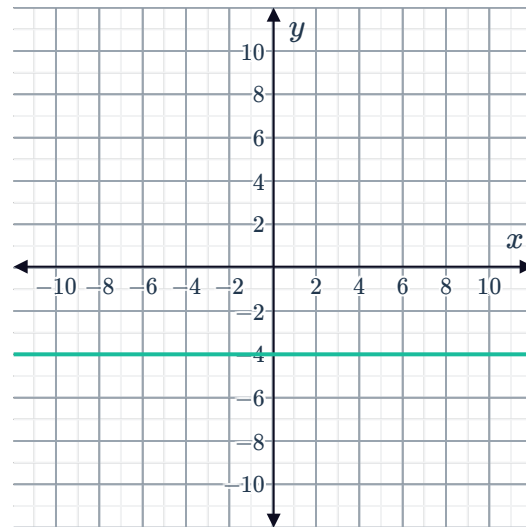
- a** State the  $x$ -value of the  $x$ -intercept of the line.
- b** How many  $y$ -intercepts does the line have?



29 Consider the graphed line.

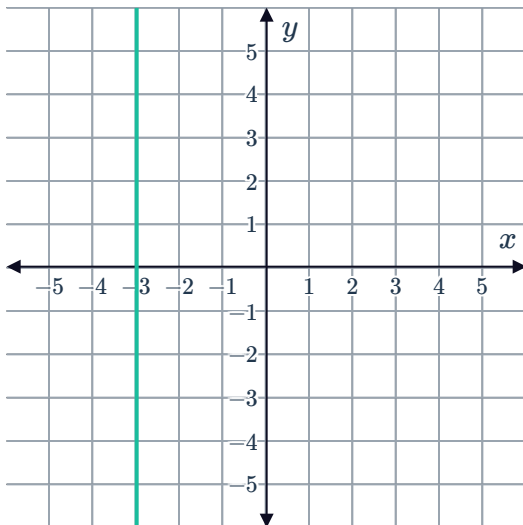


- Determine the  $y$ -value of the  $y$ -intercept of the line.
- How many  $x$ -intercepts does the line have?

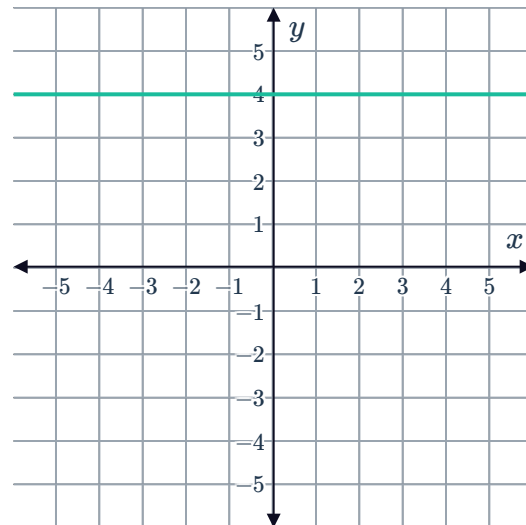


30 Determine the equation of the following graphed lines.

a



b



31 Consider the table of values shown.  
Find the equation of the line which passes through these points.

|     |      |      |      |      |
|-----|------|------|------|------|
| $x$ | -4   | -2   | 0    | 2    |
| $y$ | -4.5 | -4.5 | -4.5 | -4.5 |

32 The table shows some points on the line  $y = 0$ .  
The line  $y = 0$  represents which axis?

|     |      |      |     |     |
|-----|------|------|-----|-----|
| $x$ | -4.5 | -0.5 | 3.5 | 8.5 |
| $y$ | 0    | 0    | 0   | 0   |

- 33** The table shows some points on the line  $x = 0$ .



|     |      |      |     |     |
|-----|------|------|-----|-----|
| $x$ | 0    | 0    | 0   | 0   |
| $y$ | -5.5 | -2.5 | 3.5 | 6.5 |

The line  $x = 0$  represents which axis?

- 34** Find the equation for the following:

- a** All horizontal lines that are 10 units away from the line  $y = 3$
- b** All vertical lines that are 7 units away from the line  $x = -2$

- 35** Three of the lines that enclose a square on a number plane have equations  $x = 0$ ,  $x = 2$  and  $y = -2$ .  
Find all the possible equations of the fourth line.

- 36** The cost of a taxi ride  $C$ , in dollars, is given by  $C = 2.5t + 2$ , where  $t$  is duration of trip in minutes.
- a** For every extra minute the trip takes, how much more will the trip cost?
  - b** State what the constant value 2 could represent.

## Additional questions

- 37** Describe the transformation of  $f(x)$  that is of the form:

- a**  $g(x) = f(x) + k$
- b**  $g(x) = kf(x)$
- c**  $g(x) = -f(x)$

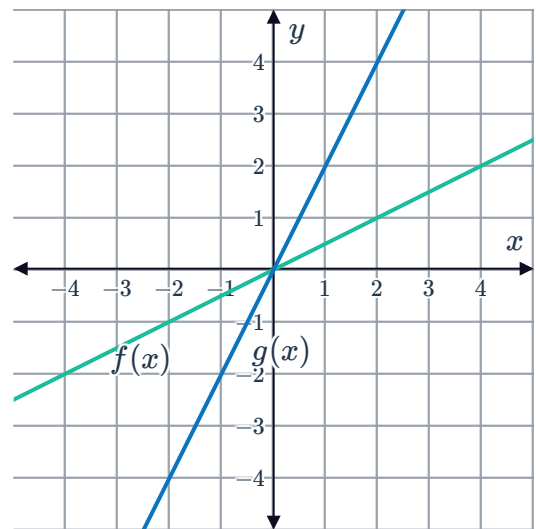
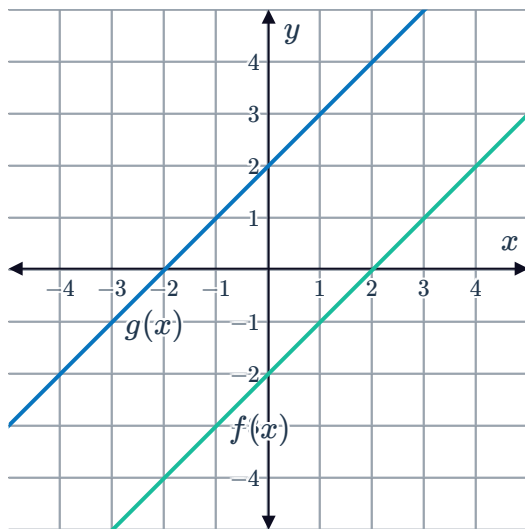
- 38** For the what values of  $k$  will the transformation from  $f(x)$  to  $g(x) = kf(x)$  involve a reflection across the  $x$ -axis?

- 39** For each of the following, describe the type of transformation from  $f(x)$  to  $g(x)$ .

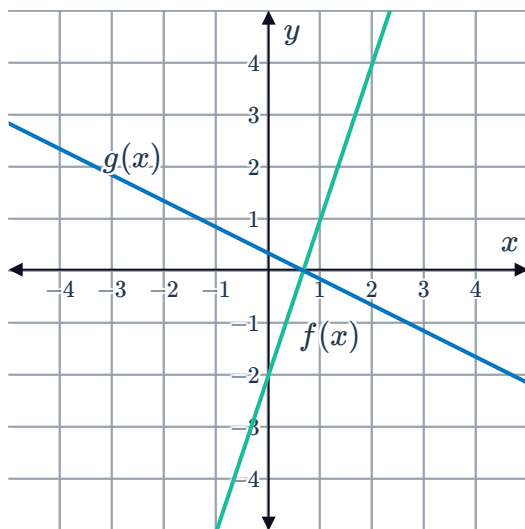
- a**  $g(x) = f(x) + 4$
- b**  $g(x) = 6f(x)$
- c**  $g(x) = \frac{1}{3}f(x)$
- d**  $g(x) = \frac{5}{12}f(x)$
- e**  $g(x) = -f(x)$
- f**  $g(x) = f(x) - 2$

- 40** For each of the graphs, describe the type of transformation from  $f(x)$  to  $g(x)$ .

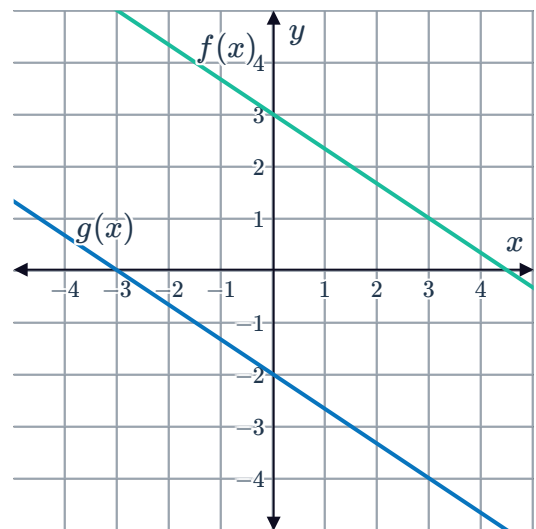
a



c

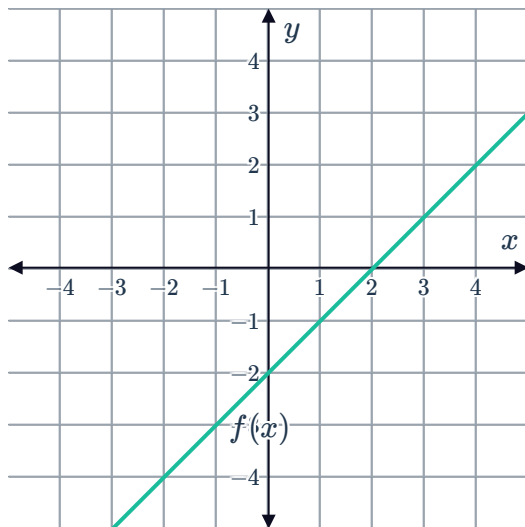


d

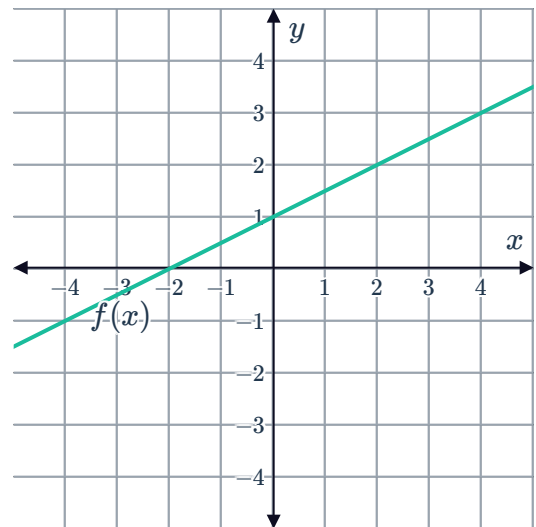


41 For each graph of  $f(x)$ , draw the graph of  $g(x)$  using the given transformation.

a  $g(x) = f(x) + 5$



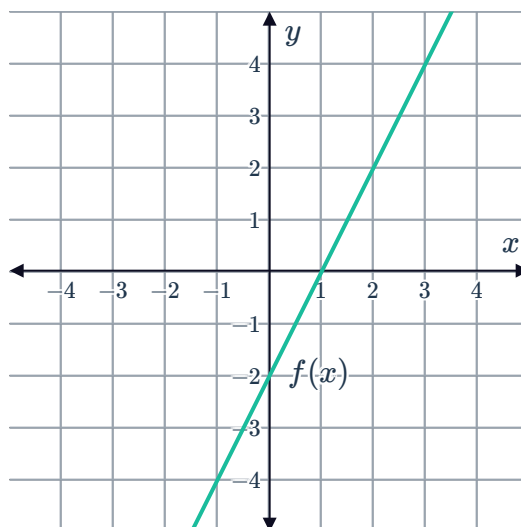
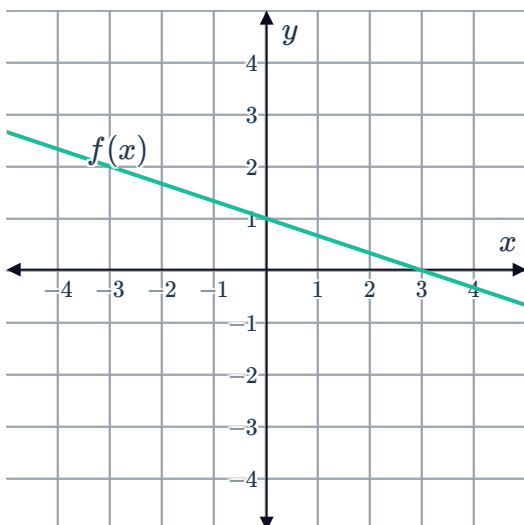
b  $g(x) = 2f(x)$



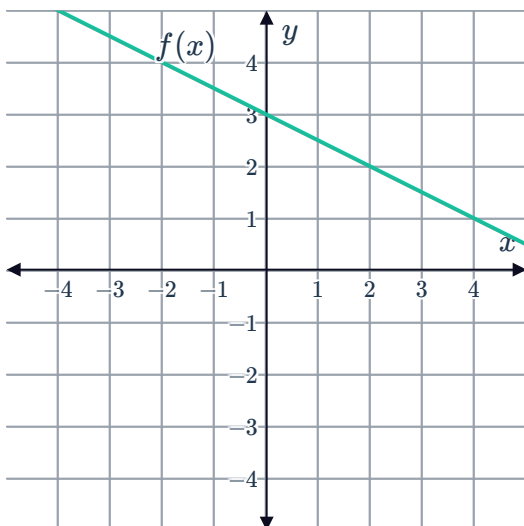
c  $g(x) = -f(x)$



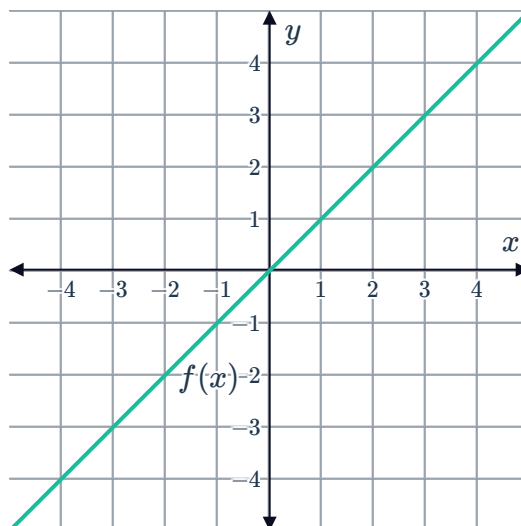
$g(x) = -\frac{1}{4}f(x)$



e  $g(x) = f(x) - 2$



f  $g(x) = f(x) + 2$



**42** For each transformation, describe how the graph of  $g(x)$  is related to the graph of  $f(x)$ .

a  $g(x) = f(x) - 8$

b  $g(x) = f(x) + \frac{1}{2}$

c  $g(x) = -3f(x)$

d  $g(x) = \frac{1}{4}f(x)$



43 Oreste draws the graph of  $g(x) = -f(x)$ , as shown.



- a Describe the error Oreste made.
- b Describe the type of graphs that  $f(x)$  could be for Oreste's error to still give him the correct answer.

